

Bundle Neural Networks for Functional Connectomes: Density-Dependent Aggregation and Cross-Site Prediction

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Abstract

Functional connectomes have an obvious graph structure, yet neighborhood aggregation can harm prediction when every node already contains a global connectivity profile. We examined whether learned orthogonal bundle transport changes this density-dependent behavior. The analysis included 754 technically eligible ABIDE-I participants from 18 sites. A single neural backbone was paired with identity propagation, a learned-local capacity control, GCN aggregation, trivial-bundle diffusion, or learned BuNN transport at positive edge densities of 0, 1, 5, 10, and 20 percent. We used nested held-out-site validation, five final seeds, regularized non-graph baselines, and representation diagnostics computed in a common frame. The primary BuNN-minus-GCN performance-curve contrast was -0.00958 (95% site-bootstrap CI [-0.03665, 0.01146]; exact sign-flip $p = 0.524$). Relative to the connectome elastic net, the BuNN contrast was -0.05516 (95% CI [-0.08297, -0.02752]). The matched-anchor effective-rank contrast was -0.00719 (95% CI [-0.01310, -0.00105]), which ran counter to the proposed preservation advantage. Site weighting and training seed affected the small BuNN-GCN difference. For this ABIDE-I pipeline, learned bundle transport provided no detected predictive or representation-preservation advantage. This finding concerns a computational operator under a specified analysis; it does not establish a biological bundle geometry or a general result for BuNNs.

1 Introduction

Resting-state functional connectomes summarize statistical associations among regional blood-oxygen-level-dependent time series. Representing brain regions as nodes and their functional associations as edges makes graph learning seem like a natural choice. The usual node feature, however, is an entire row of the connectivity matrix. Each node begins with information about the whole connectome. Aggregating neighboring rows may blur useful differences rather than supply missing context.

A recent benchmark across several datasets found that classical models and non-graph neural networks can match or outperform graph deep-learning models on functional-connectome tasks. Its ablations identified message aggregation as one possible source of degradation [Han et al., 2026]. A graph representation alone is not enough to justify propagation. The operator must earn its place against identity propagation and strong regularized baselines evaluated on the same splits.

BuNNs first transport features between node-specific frames using learned orthogonal maps and then diffuse the aligned features [Bamberger et al., 2025]. Theory and synthetic experiments suggest that this construction can resist representation collapse more effectively than ordinary graph diffusion. Neural-sheaf models provide related evidence that non-trivial transport changes diffusion behavior [Bodnar et al., 2022]. Dense functional connectomes with globally informed node features present a different test.

The project initially treated negative functional correlations as a possible form of heterophilic message passing. That interpretation did not survive formalization. Correlation sign does not identify node-label heterophily, neural excitation or inhibition, anatomy, or causal flow, and it can change with preprocessing choices such as global-signal regression [Murphy and Fox, 2017]. We kept the dataset, connectome representation, BuNN architecture, and original aggregation problem, but shifted the hypothesis to the behavior of the propagation operator itself.

We ask whether learned bundle transport yields a better held-out-site performance-density curve than GCN aggregation. Any advantage must also exceed identity propagation, the learned-local capacity control, and a regularized connectome baseline. We separately test whether the predictive pattern co-occurs with representation preservation measured in a common frame. The scope is one pre-specified ABIDE-I pipeline and one architectural hypothesis; biological geometry and clinical diagnosis lie outside it.

2 Related work

Connectome prediction. ABIDE-I aggregated resting-state fMRI and phenotypic data across international sites to support large-scale autism research [Di Martino et al., 2014]. The Preprocessed Connectomes Project subsequently released derivatives from several pipelines, including C-PAC [Preprocessed Connectomes Project, 2026]. Multi-site prediction is difficult because acquisition, motion, demographics, and pre-processing vary by site. Holding out whole sites tests a stricter form of generalization than a random participant split.

Graph aggregation. GCNs apply normalized neighborhood aggregation followed by learned feature transformation [Kipf and Welling, 2017]. Repeated diffusion can contract node representations, although the practical effect depends on architecture, features, graph construction, and task. In functional connectomes, the problem matters because a connectivity-row node feature already describes the whole graph. The findings of Han et al. [2026] make a no-propagation control essential. A comparison among GNN families alone cannot show that aggregation helped.

Learned transport. Neural sheaf diffusion assigns vector spaces and learned maps to graph incidences, allowing non-trivial transport before aggregation [Bodnar et al., 2022]. BuNN specializes this idea to flat vector bundles with orthogonal node maps and diffusion-based propagation [Bamberger et al., 2025]. Raw embeddings from different node frames cannot be compared directly. Collapse diagnostics must either place them in a common frame or use a gauge-invariant quantity. We also include a learned-local control so that node-wise map capacity is not mistaken for an effect of inter-node transport.

3 Methods

3.1 Study design

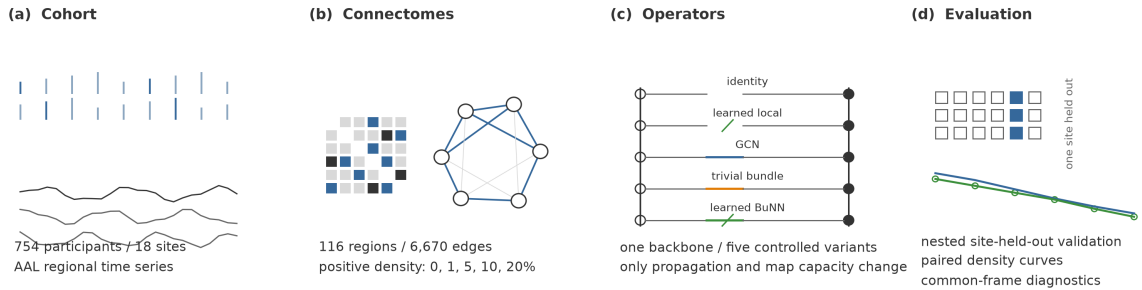


Figure 1: Study design. Preprocessing, model selection, and hyperparameter selection used training sites only. Each outer site remained untouched until its final evaluation.

The experiment isolates the propagation operator while holding the surrounding pipeline fixed (figure 1). We completed and audited the classical baselines before fitting the neural models. Five propagation or capacity-control variants then shared one neural backbone. They used identical outer sites, grouped inner folds, candidate budgets, stopping rules, and final seeds. Before examining performance, we verified the complete neural artifact and froze the confirmatory analysis code and decision rules.

3.2 Dataset and technical quality control

We used the ABIDE-I C-PAC derivatives released by the Preprocessed Connectomes Project [Di Martino et al., 2014, Preprocessed Connectomes Project, 2026]. The source was the band-pass-filtered AAL ROI time-series derivative without global-signal regression. The AAL atlas defines macroscopic anatomical regions in standardized space [Tzourio-Mazoyer et al., 2002]. Eligibility required a valid header, finite numeric values, non-zero variance in every ROI, and a finite resulting connectome. Of 769 downloaded files, 15 failed the

zero-variance check. The retained cohort contained 754 participants (371 ASD and 383 controls) from 18 evaluable sites. These checks establish technical eligibility, not population representativeness.

Table 1: Cohort retained after technical quality control and resulting connectome dimensions.

Item	Value
Downloaded AAL time-series files	769
Technical exclusions	15
Retained participants	754
ASD / control	371 / 383
Held-out sites	18
ROIs / lower-triangle features	116 / 6,670

3.3 Connectomes, node features, and graphs

For each participant, we calculated Pearson correlations between every pair of 116 ROI time series and applied Fisher’s r -to- z transformation. We set the diagonal to zero. The 6,670 values in the lower triangle of the finite, symmetric matrix formed the classical feature vector. For the neural models, each node received its complete connectivity row as a 116-dimensional feature. The node inputs were thus globally informed before any message passing occurred.

At nonzero densities, we retained the strongest positive off-diagonal associations symmetrically at 1, 5, 10, and 20 percent. Density zero contained only the matched identity or learned-local anchor. The exclusion of negative associations was a pre-specified computational choice, not a biological interpretation.

3.4 Classical baselines

We fit three logistic baselines: covariates alone, vectorized connectomes with elastic-net regularization, and connectomes combined with covariates. Elastic-net regularization combines ℓ_1 and ℓ_2 penalties and is appropriate for correlated high-dimensional predictors [Zou and Hastie, 2005]. Covariates were age, sex, mean framewise displacement, and usable scan length. Scaling, feature processing, and tuning used training sites only.

3.5 Shared neural backbone and operators

All variants shared the encoder, hidden width, depth, nonlinearities, normalization, dropout, pooling, classifier, optimizer, training schedule, and tuning budget. Only propagation and the capacity associated with learned maps changed:

1. **Identity:** no inter-node propagation.
2. **Learned-local:** the BuNN map generator and node-wise updates, but no inter-node diffusion, controlling for additional learned capacity.
3. **GCN:** normalized neighborhood aggregation [Kipf and Welling, 2017].
4. **Trivial bundle:** the same diffusion construction with identity transport maps.
5. **Learned BuNN:** learned orthogonal node maps followed by bundle diffusion [Bamberger et al., 2025].

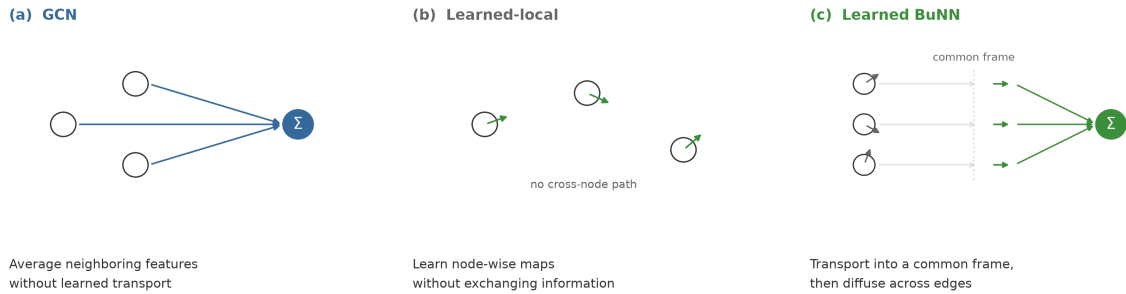


Figure 2: Schematic distinction among GCN aggregation, the learned-local capacity control, and learned bundle transport. The drawing describes a computational operation only; it does not represent biological signal flow.

3.6 Nested held-out-site evaluation

Each of the 18 sites served once as the outer test site. Four grouped folds within the remaining sites selected among four equal-budget AdamW candidates using two tuning seeds. We refit the selected configuration with five fixed final seeds. Test-site data did not enter scaling, tuning, early stopping, or configuration selection. Equal-site mean balanced accuracy was the primary predictive endpoint. Pooled balanced accuracy and AUROC were secondary.

3.7 Density curves and representation diagnostics

The primary predictive estimand was the normalized trapezoidal area under the balanced-accuracy curve across the full density grid. GCN and trivial-bundle curves used identity at density zero. The learned-BuNN curve used the parameter-matched learned-local anchor. Evaluating the entire curve prevented the analysis from selecting a favorable density after the fact.

After every propagation layer, we mapped embeddings into a common learned frame before comparing nodes. Normalized effective rank at layer 2 was the primary representation endpoint. Secondary measures were normalized dispersion, mean pairwise cosine similarity, invariant edge-transport distance, and the change in rank from encoder to layer 2. Prediction and representation were analyzed as co-occurring outcomes, without a mediation or causal claim.

3.8 Statistical analysis and decision rule

We averaged the five final seeds within each site and configuration before site-level inference. Paired percentile intervals came from 10,000 bootstrap resamples of the 18 held-out sites. Curve contrasts also received exact two-sided paired sign-flip tests. Density-specific tests were secondary and Holm-corrected within each named contrast family.

The positive-result rule required three conditions: stronger representation preservation than GCN, better held-out prediction than GCN and the no-propagation controls, and performance above the strongest regularized baseline. We reported a confidence interval containing zero as no detected advantage. We ruled out a practically meaningful gain only when the upper bound excluded the pre-specified +0.03 balanced-accuracy margin.

3.9 Run integrity and reproducibility

Long runs wrote atomic checkpoints after each site and recorded immutable hashes for the data, configuration, and source. Deterministic site ownership, resume validation, and independent completion audits guarded the distributed run. The accepted neural artifact covered every outer site and passed checks for coverage, recomputation, diagnostics, runtime, and warnings before unblinding. Manuscript values are generated from a hash-bound result snapshot instead of being copied by hand.

4 Results

4.1 Cohort and classical baselines

After technical quality control, 754 participants remained across 18 held-out sites (table 1). The connectome elastic net had the highest equal-site balanced accuracy among the classical models (0.6401). The corresponding values were 0.6336 for connectomes plus covariates and 0.5652 for covariates alone (table 2). These cross-site results describe computational prediction, not clinical diagnostic performance.

Table 2: Classical held-out-site baselines. Equal-site balanced accuracy is the primary summary.

Model	Equal-site BA	Pooled BA	Pooled AUROC
Covariates-only logistic regression	0.5652	0.5629	0.5973
Connectome elastic net	0.6401	0.6495	0.6794
Connectome + covariates elastic net	0.6336	0.6428	0.6773

4.2 Predictive density curves

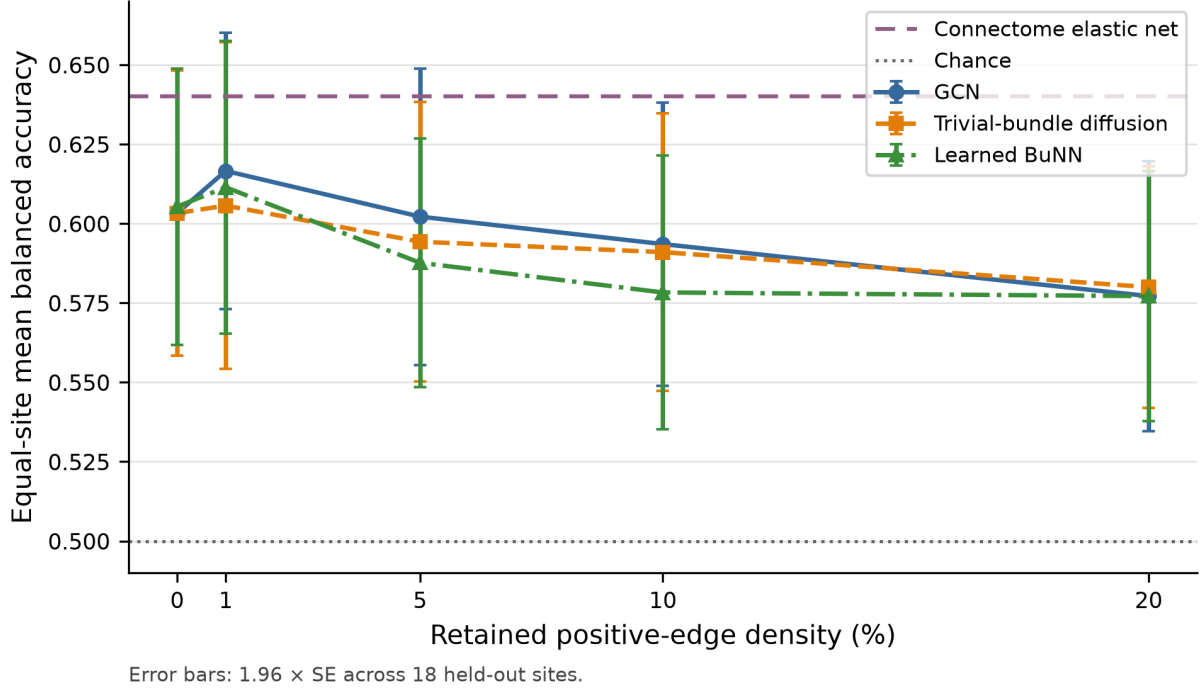


Figure 3: Equal-site mean held-out balanced accuracy across the pre-specified positive-edge density grid. Error bars show 1.96 standard errors across sites for visualization; confirmatory inference used paired site bootstrap and exact sign-flip tests on the locked curve estimand.

All three diffusion curves declined as density increased (figure 3). The normalized curve-area difference for learned BuNN minus GCN was -0.00958 (95% site-bootstrap CI $[-0.03665, 0.01146]$; exact $p = 0.524$). Because the interval crossed zero, we detected no predictive advantage. The upper bound also fell below the pre-specified $+0.03$ margin. This rules out a gain of that size for the chosen estimand and pipeline, but it neither establishes universal equivalence nor excludes smaller gains.

Against the site-matched connectome elastic net, the learned-BuNN curve difference was -0.05516 (95% CI $[-0.08297, -0.02752]$; exact $p = 0.0018$). Comparisons with identity propagation and the learned-local control also failed the positive-result rule (table 3).

Table 3: Pre-specified predictive contrasts. Differences are BuNN minus the named comparator.

Contrast	Estimate	95% site-bootstrap CI	Exact p
BuNN curve - GCN curve	-0.0096	$[-0.0366, 0.0115]$	0.5237
BuNN curve - trivial-bundle curve	-0.0062	$[-0.0293, 0.0108]$	0.6813
BuNN nonzero mean - learned-local	-0.0167	$[-0.0347, 0.0009]$	0.0958
BuNN nonzero mean - identity	-0.0146	$[-0.0320, 0.0018]$	0.1155
BuNN curve - connectome elastic net	-0.0552	$[-0.0830, -0.0275]$	0.0018

4.3 Common-frame representation diagnostics

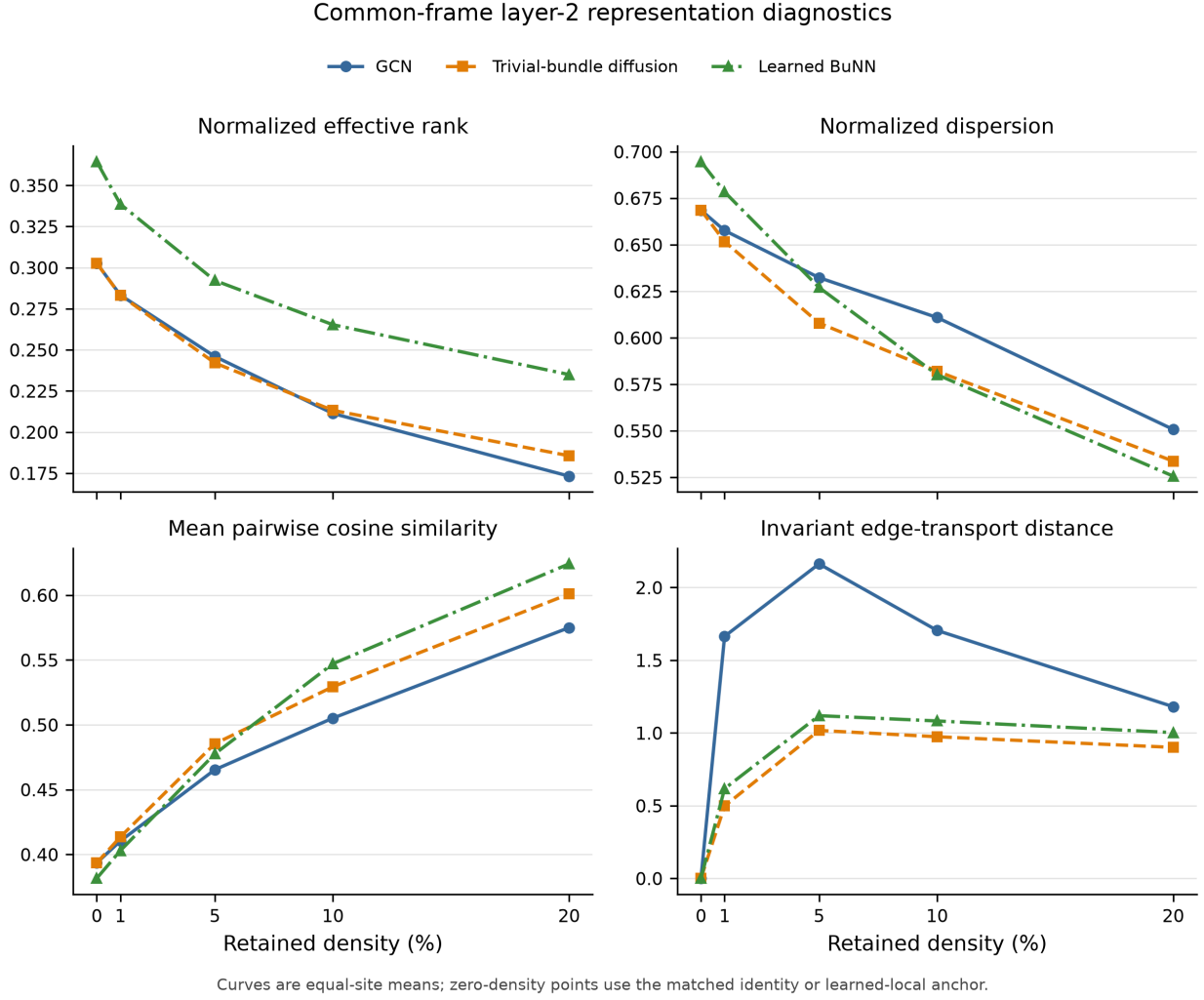


Figure 4: Equal-site means for common-frame layer-2 representation diagnostics. Zero-density points use the matched identity or learned-local anchor. Raw unaligned BuNN node embeddings were not compared.

The matched-anchor normalized effective-rank contrast was -0.00719 (95% CI $[-0.01310, -0.00105]$), the opposite of the proposed BuNN preservation advantage. Although the raw BuNN rank curve was higher, its learned-local anchor already included the extra map-generator capacity. The matched contrast did not support attributing the raw difference to inter-node bundle transport. Dispersion and cosine similarity pointed in the same direction (figure 4). None of these diagnostics identifies a biological mechanism.

4.4 Site, seed, and weighting sensitivity

Only 1 of the 18 leave-one-site estimates was positive, and no favorable exclusion interval lay above zero (Supplementary Figure S1). Of the five final seeds, 2 point estimates favored BuNN, but 0 favorable intervals excluded zero. 1 unfavorable seed interval lay below zero (Supplementary Figure S2). Participant weighting reversed the sign of the small BuNN-GCN point estimate; the equal-site mean and median remained negative. The pre-specified confirmatory estimand used equal-site weighting.

4.5 Observed execution cost

Learned BuNN used 8,529 parameters, compared with 5,889 for GCN. Its recorded fits required 31.14 total fit-hours, while GCN required 15.09. Equal-site curve balanced accuracy was lower for BuNN (0.5849 versus 0.5945). The timings apply to this implementation on an A10G GPU and should not be read as universal complexity estimates.

Table 4: Observed execution cost and predictive curve summary for the three diffusion operators. Runtime is implementation- and hardware-specific.

Operator	Params.	Fit h	s/fit	Peak GiB	Curve BA
GCN	5,889	15.09	20.39	0.373	0.5945
Trivial-bundle diffusion	5,889	25.99	35.12	0.377	0.5911
Learned BuNN	8,529	31.14	42.08	0.378	0.5849

5 Discussion

The proposed BuNN advantage did not carry over to this ABIDE-I functional-connectome pipeline. Learned bundle transport did not improve the primary performance-density curve over GCN, exceed the strongest regularized connectome baseline, or preserve matched-anchor effective rank more strongly. Because the backbone, inputs, tuning budget, and capacity controls were held fixed, these negative findings speak directly to propagation rather than to differences elsewhere in the models.

5.1 Why transport may not help in this setting

This task differs from the settings that motivate bundle transport in two important ways. Functional-connectome graphs remain dense after thresholding, and each node carries its full connectivity row. Every node already has global information before propagation. Diffusion may be averaging informative global profiles rather than delivering missing long-range context. The declining density curves are consistent with aggregation degradation, but they cannot by themselves establish a general over-smoothing mechanism.

5.2 Importance of the learned-local control

BuNN adds parameters through its learned map generator. Comparing it with GCN alone could mistake added node-wise capacity for an effect of transport. The learned-local control kept that capacity but removed inter-node diffusion. Its high raw effective rank shows how an unaligned or unmatched rank curve could give a misleading impression of preservation. Anchoring each operator to its matched density-zero control isolates the change introduced by inter-node transport.

5.3 Heterogeneity and uncertainty

The primary BuNN-GCN interval crossed zero, and participant weighting reversed the point-estimate sign. Leaving out individual sites and changing final seeds also altered the magnitude, and sometimes the direction, of this small contrast. The sensitivity analyses show that the average difference is not stable relative to site and optimization variation. By comparison, the BuNN-versus-elastic-net contrast and the matched-anchor effective-rank result kept their direction under every leave-one-site exclusion.

5.4 Limitations

The result is conditional on ABIDE-I, C-PAC filtered data without global-signal regression, the AAL-116 atlas, Pearson correlations with Fisher transformation, positive-edge thresholding, connectivity-profile node features, the fixed density grid, the shared backbone, and the equal tuning budget. Functional correlations do not identify anatomy or causal flow. No locked external cohort was available, and site sample sizes differed substantially. Runtime depends on both implementation and hardware. Representation diagnostics describe computational behavior, not a causal link between representation change and prediction.

5.5 Future work

A useful next step is to lock the full ABIDE-I pipeline and evaluate it on a preprocessing-compatible ABIDE-II cohort. Separate pre-registered studies could examine global-signal regression, negative-edge handling, a different atlas, or node features that do not encode the full connectome. Keeping those questions separate would protect the present confirmatory result from post-hoc dataset or diagnosis selection.

6 Conclusion

In this pre-specified ABIDE-I pipeline, learned orthogonal bundle transport did not improve the held-out-site performance-density curve over GCN. It also failed to surpass identity propagation or the learned-local control and remained below the regularized connectome baseline. Gauge-aware, matched-anchor diagnostics offered no evidence for the proposed representation-preservation advantage. Bundle transport changed the operator and increased both model capacity and runtime, but we detected no predictive benefit in this dense connectome setting with globally informed node features.

Data and code availability

This analysis uses public ABIDE-I derivatives from the Preprocessed Connectomes Project. The reproducibility package includes the data manifest, technical exclusions, site splits, configurations, seeds, source hashes, analysis scripts, and aggregate result artifacts. Participant-linked audit records are kept separate from the public release package.

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